Minimum Wage Shocks on Unemployment: Evidence from California's 2021 Policy

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Abstract

This study examines the labor market effects of California's January 2021 minimum wage increase. Using county-level monthly unemployment data from California and New York between 2018 and 2025, we estimate a Difference-in-Differences model with county and time fixed effects. To account for dynamic persistence in unemployment, we include a one-period lag of the outcome variable. We find that, after the policy, unemployment in California counties rose by 2.3% relative to New York, suggesting a moderate but statistically significant policy impact. The effect remains statistically robust across model specifications. Fixed effect estimates reveal persistent differences across counties and capture economy-wide shocks such as the COVID-19 outbreak. We further find no significant treatment effect heterogeneity by counties' baseline unemployment levels. Taken together, our findings suggest that binding wage floors can raise unemployment in competitive labor markets, consistent with classic economic theory.

Keywords: Minimum Wage, Unemployment, Difference-in-Differences, Labor Policy

JEL Codes: J08, J38, E24, C21

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1 Question

In this study, we focus on the causal impact of California's 2021 minimum wage increase on county-level unemployment and pose the following research questions: Did the policy lead to a significant rise in unemployment during the post-COVID recovery period? Did the effect vary across counties with different baseline unemployment conditions?

2 Motivation

Minimum wage policy has long been at the center of debates in labor economics, with particular attention to its potential trade-off between wage growth and employment. Policymakers often advocate minimum wage increases as a tool to improve income equity and reduce working poverty, yet classical economic theory warns of potential job losses, especially in competitive labor markets where wages are assumed to reflect marginal productivity. This tension between efficiency and equity raises a fundamental question: Do minimum wage hikes help or hurt the very workers they aim to protect?

On January 1, 2021, the state of California raised the minimum wage from \$12 to \$13 per hour for employers with 25 or fewer employees, and from \$13 to \$14 per hour for employers with 26 or more employees. This policy provides an opportunity to empirically evaluate the labor market consequences of a binding wage floor. Given California's size, economic diversity, and relatively high cost of living, its labor market response may also serve as a leading indicator for similar policies in other states or at the federal level.

From a theoretical standpoint, the canonical competitive model predicts that a binding minimum wage leads to excess supply of labor, i.e., unemployment, particularly among low-skilled workers. On the other hand, alternative frameworks such as monopsony models suggest that a moderate increase in the minimum wage could improve both wages and employment by countering employer wage-setting power. Which of these forces dominates in practice remains an empirical question, and may vary across local labor markets depending on baseline conditions.

Beyond its theoretical significance, understanding the effects of minimum wage changes is crucial for policy design. If the policy raises unemployment in some regions but not others, or among some demographic groups but not others, targeted or complementary policies may be needed. Moreover, the COVID-19 pandemic introduced unprecedented shocks to labor markets, and measuring policy effects in such a volatile period provides further insights into the resilience and adaptability of local economies.

This study contributes to the literature by leveraging high-frequency county-level unemployment data from both California (treatment group) and New York over an extended time frame (2018–2025), allowing for rigorous causal inference using a Difference-in-Differences framework. We also explore treatment effect heterogeneity by baseline unemployment conditions to test whether pre-existing labor market tightness shapes a region's response to wage mandates.

This study is motivated by a timely and policy-relevant question with theoretical ambiguity and real-world consequences. By providing credible evidence on the labor market effects of minimum wage policy, our findings aim to inform ongoing state and federal policy discussions.

3 Literature Review

3.1 Theoretical Foundation

A theoretical foundation for analyzing the effects of minimum wage policy comes from Stigler (1946), who models how wage floors operate in competitive labor markets. His theory predicts that when the minimum wage, denoted as $w_{\min}$, is set above the market-clearing wage w^*, it creates a mismatch between labor supply and labor demand.

Labor demand is defined as $L_D = D(w)$, where L_D is the quantity of labor demanded and w is the wage level. The function $D(w)$ is downward-sloping, meaning that $D'(w) < 0$; as wages rise, firms demand less labor. Labor supply is defined as $L_S = S(w)$, where L_S is the quantity of labor supplied. The function $S(w)$ is upward-sloping, meaning that $S'(w) > 0$; as wages rise, more individuals are willing to work. When the wage floor $w_{\min}$ exceeds the equilibrium wage,

the amount of labor supplied exceeds the amount of labor demanded. This leads to involuntary unemployment, which is defined as the difference between labor supply and labor demand at the minimum wage level: $U = S(w_{\min}) - D(w_{\min})$, where U is the level of unemployment created by the policy. This theoretical prediction supports our hypothesis that California's 2021 minimum wage increase could have led to higher unemployment, particularly among lower-productivity or low-wage workers.

To ensure that Stigler's assumptions are applicable to our empirical setting, we check that both California and New York exhibit characteristics of competitive labor markets. These include large and diverse labor forces, high rates of worker mobility, and significant competition among employers. These features justify the use of Stigler's competitive framework and enhance the validity of our Difference-in-Differences research design.

3.2 Covariates Control

Cengiz et al. (2019) use data from the Current Population Survey (CPS), covering the years 1979 to 2016, to examine the impact of 138 state-level minimum wage increases across the U.S. Their empirical strategy includes a bunching estimator that isolates changes in employment by comparing "missing" jobs below the new minimum wage to "excess" jobs that appear at or slightly above it.

To ensure robustness, the authors control for a range of covariates, including demographic factors (age, gender, race, education) and macroeconomic trends such as national recessions and booms. They incorporate wage-bin-by-state and period fixed effects, and test multiple specifications with linear and quadratic time trends to address potential omitted variable bias. These controls strengthen the causal interpretation of their findings.

Their main conclusion is that minimum wage increases do not result in significant job losses for low-wage workers. However, their findings differ from ours for two key reasons. First, Cengiz et al.'s sample spans both recessionary and expansionary periods, whereas our study focuses exclusively on the post-COVID recovery—a time when many businesses were already financially

strained and more likely to cut jobs. Second, while their analysis targets low-wage jobs nationwide, our Difference-in-Differences design examines total unemployment in California and New York. This broader focus at the state level may reveal stronger aggregate effects not captured in studies limited to the low-wage sector.

3.3 Robust Check Extensions

Angrist and Pischke (2009) emphasize the importance of verifying the parallel trends assumption in DID designs. In *Mostly Harmless Econometrics*, they recommend practical strategies such as including state-specific linear time trends or interacting treatment status with time indicators to test for pre-treatment differences. If the interaction terms are statistically insignificant, the parallel trends assumption is more likely to hold. In line with this approach, we estimate a model that allows for differential pre-trends between California and New York. This strategy helps rule out the possibility that post-treatment differences are driven by pre-existing divergent trends rather than by the policy itself.

Card and Krueger (1994), in their study of the 1992 New Jersey minimum wage increase, perform robustness checks using alternative outcome variables that are unlikely to be directly affected by minimum wage changes. For example, they analyze store-level characteristics such as hours of operation, the number of cash registers, and employee meal programs. The absence of significant effects on these outcomes enhances the credibility of their main findings. Following this logic, we conduct similar robustness checks using alternative labor market indicators (e.g., employment in unrelated sectors) to verify that the observed changes in unemployment are plausibly attributable to the minimum wage policy rather than to unrelated economic fluctuations.

To address concerns related to policy endogeneity, Besley and Case (2000) propose using placebo tests in panel data settings. In their study of workers' compensation policies, they implement placebo DID models using periods in which no policy changes occurred, checking whether the empirical strategy falsely detects treatment effects. Inspired by this method, we conduct placebo tests by assigning false treatment dates to pre-policy periods (e.g., comparing 2019 to 2020) to

ensure that no significant changes emerge prior to the actual policy implementation. The absence of effects during these placebo periods strengthens the credibility of our identification strategy.

4 Data

We scraped county-level data for both California and New York from the Bureau of Labor Statistics (BLS) using a combination of manual download and an automated script. The variables retrieved from the dataset were the number of unemployed individuals in each county. To pre-process the dataset, we merged multi-year dataset and validated consistency for California and New York, excluding data where there wasn't information for both datasets. The time frame of the data analyzed was from January 2018 to January 2025, with monthly resolution in the data.

Table 1 presents summary statistics for county-level unemployment in California and New York. On average, counties in California recorded significantly higher unemployment counts than those in New York (19,005 vs. 8,239). The standard deviation in California was more than twice as large, indicating greater variation in unemployment across counties. California also exhibited a much higher maximum unemployment level (935,509) compared to New York (257,898), while the minimum values were similarly low in both states. These descriptive patterns highlight the importance of accounting for baseline differences in unemployment when assessing policy impacts. We do not include time-varying county-level covariates such as demographic composition or industrial structure. This decision reflects the lack of consistent monthly data at the county level. Nonetheless, our use of county and time fixed effects helps address potential omitted variable bias by controlling for unobserved heterogeneity across counties and over time.

Table 1: Summary Statistics for County-Level Unemployment

State	Mean	Median	Std. Dev.	Min	Max
California	19005	4609	51979	19	935509
New York	8239	1706	19381	61	257898

Looking at the autocorrelation and partial autocorrelation functions, the data shows an AR(1) process, showing that controlling for the first-order lag of unemployment is necessary to account for dynamic persistence and focus on the shock effect.

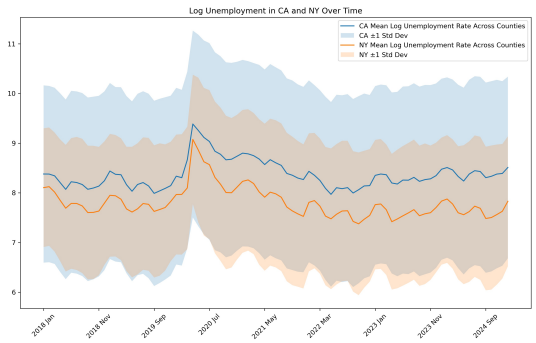


Figure 1: Log(Unemployment) Over Time

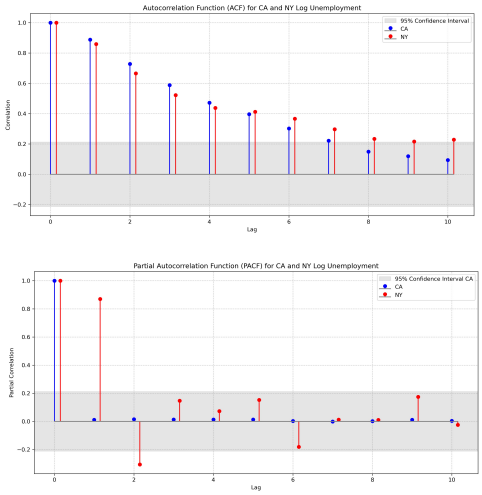


Figure 2: ACF and PACF Plots

We also evaluate the effectiveness of other data sources, especially those from the American Community Survey and the Current Population Survey.

The American Community Survey, which is produced by the US Census Bureau, annually gathers information that is only collected in the long form of the US Census, which happens once every 10 years. The survey is sent to 3.5 million household each year, and collects data based on labor force participation, employment, and other such statistics.

The Current Population Survey, jointly sponsored by both the US Census Bureau and the Bureau of Labor Statistics, surveys 60,000 US households every month and provides estimates of the unemployment rate and other related statistics. A readable summary is provided monthly, and more granular data can be used by researchers to perform further analyses.

Both of these are surveys, and while they do record richer levels of data regarding employment, there could be significant selection bias with those who are unhoused not receiving the survey, or not having the time to respond to it. Therefore, we use the BLS data because of its more comprehensive nature with the individuals covered. With over 3,000 counties in the United States,

both of these surveys are unlikely to have a survey size large enough for it to be robust enough for the analyses we are performing.

5 Econometric Strategy

5.1 Identification strategy

Under the Rubin's Causal Model, our objective is to estimate the Average Treatment Effect on the Treated (ATT), which in this context corresponds to the average impact of California's minimum wage policy on treated counties in the post-policy period. Formally, the ATT is defined as:

$$\text{ATT} = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid D_i^{\text{treatment}} = 1, t \geq \text{Jan 2021}]$$

Since $Y_{it}(0)$ is unobservable for treated counties after the policy, we use control counties in New York to approximate this counterfactual. The identification of ATT relies on the parallel trends assumption, which, in the context of our study, implies that had the minimum wage policy not been implemented in California, the unemployment trends in California counties would have evolved similarly to those in New York counties. This parallel trends assumption is critical to ensuring that any post-treatment divergence can be attributed to California's 2021 minimum wage increase, rather than to pre-existing differences or unrelated shocks.

To assess the plausibility of this assumption, we use data from 2018 to 2020 and conduct both graphical and statistical diagnostics. In particular, we compare the trajectories of log unemployment across the two states prior to treatment and estimate models that allow for state-specific time trends. The absence of significant differential pre-trends supports the validity of our identification strategy.

Figure 3 plots the log of monthly total unemployment for California and New York before treatment. Both series exhibit relatively stable trends and no strong divergence prior to 2020. The figure suggests that the pre-treatment trajectories are broadly similar. We also observe a sharp spike in March 2020, which reflects the nationwide economic shock caused by the onset of COVID-19

lockdown policies.

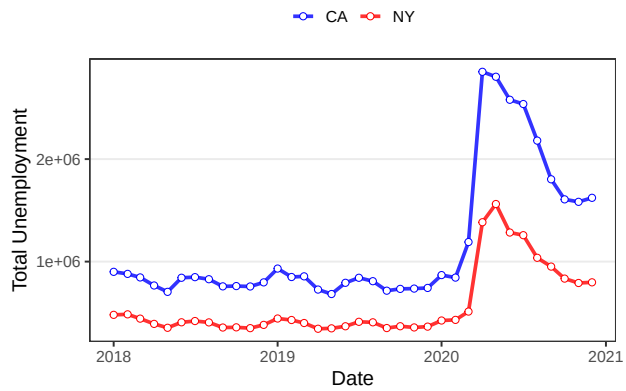


Figure 3: Pre-Treatment Trends

5.2 Regression Specification

We evaluate the effect of the policy intervention using three DID models, all centered around a canonical specification that compares unemployment trends across treated and untreated groups before and after policy implementation. The benchmark model is the simple DID regression:

$$\log(Y_{it}) = \beta_0 + \beta_1 D_i^{\text{treatment}} + \beta_2 D_t^{\text{post}} + \beta_3 D_i^{\text{treatment}} D_t^{\text{post}} + u_{it}$$

In this equation, $\log(Y_{it})$ is the natural logarithm of monthly unemployment in county i at time t, $D_i^{\text{treatment}}$ is a treatment indicator equal to 1 if the county is in California and 0 otherwise, and D_t^{post} is a post-treatment indicator equal to 1 for months after January 2021. The interaction term $D_i^{\text{treatment}} D_t^{\text{post}}$ captures the average treatment effect of the policy, denoted by β_3.

Building on this baseline, we propose two enriched models:

1. Two-Way Fixed Effects Model (TWFE): This model adds county and time fixed effects to account for time-invariant heterogeneity across counties and aggregate shocks over time. The specification is:

$$\log(Y_{it}) = \beta_0 + \beta_1 D_i^{\text{treatment}} + \beta_2 D_t^{\text{post}} + \beta_3 D_i^{\text{treatment}} D_t^{\text{post}} + \gamma_i + \delta_t + u_{it}$$

where γ_i are county fixed effects and δ_t are time fixed effects.

2. TWFE with Lagged Response: To account for potential serial correlation in unemployment, we introduce a lagged dependent variable:

$$\log(Y_{it}) = \beta_0 + \beta_1 D_i^{\text{treatment}} + \beta_2 D_t^{\text{post}} + \beta_3 D_i^{\text{treatment}} D_t^{\text{post}} + \rho \log(Y_{i,t-1}) + \gamma_i + \delta_t + u_{it}$$

6 Results

6.1 Main Regression

Table 2 presents the estimation results across all three models. In the simple DID model, the estimated coefficient on the treatment indicator is $\hat{\beta}_3 = 0.213$, indicating a statistically significant increase in unemployment in California counties relative to New York after the policy took effect. Interpreted in percentage terms using the standard transformation $(e^{\hat{\beta}_3} - 1) \times 100\%$, this corresponds to a 23.7% relative increase in unemployment. The second model yields the same point estimate but with a much smaller standard error, suggesting greater precision after controlling for county and month fixed effects. The model also has a stronger explaining power, with R^2 rising from 0.032 to 0.993.

In the third model, we further include the one-period lag of log unemployment to account for the persistence in labor market conditions. After this adjustment, the estimated treatment effect decreases substantially to $\hat{\beta}_3 = 0.023$, which implies a relative increase of approximately 2.3%. This substantial decrease is expected: unemployment is highly persistent over time, and the lag term $\log(Y_{i,t-1})$ absorbs much of the variation that might otherwise be attributed to the policy. Nonetheless, the estimate remains statistically significant, suggesting that the policy had a genuine effect beyond the continuation of pre-existing trends.

Table 2: DID Estimates across Model Specifications

	Simple DID	**Two-Way FE**	**Two-Way FE (Lag)**
Intercept	7.962^{***}	6.273^{***}	0.921^{***}
	(0.034)	(0.020)	(0.033)
Treatment	0.440^{***}	4.301^{***}	0.639^{***}
	(0.050)	(0.021)	(0.024)
Post	-0.267^{***}	-0.183^{***}	0.107^{***}
	(0.045)	(0.019)	(0.009)
Treatment × Post	0.213^{**}	0.213^{***}	0.023^{***}
	(0.065)	(0.006)	(0.003)
log$(Y_{i,t-1})$	–	–	0.854^{***}
			(0.006)
Observations	10,200	10,200	10,080
R^2	0.0322	0.9932	0.9983
Adjusted R^2	0.0319	0.9930	0.9982
F-statistic	113.2	7118.6	27761.4

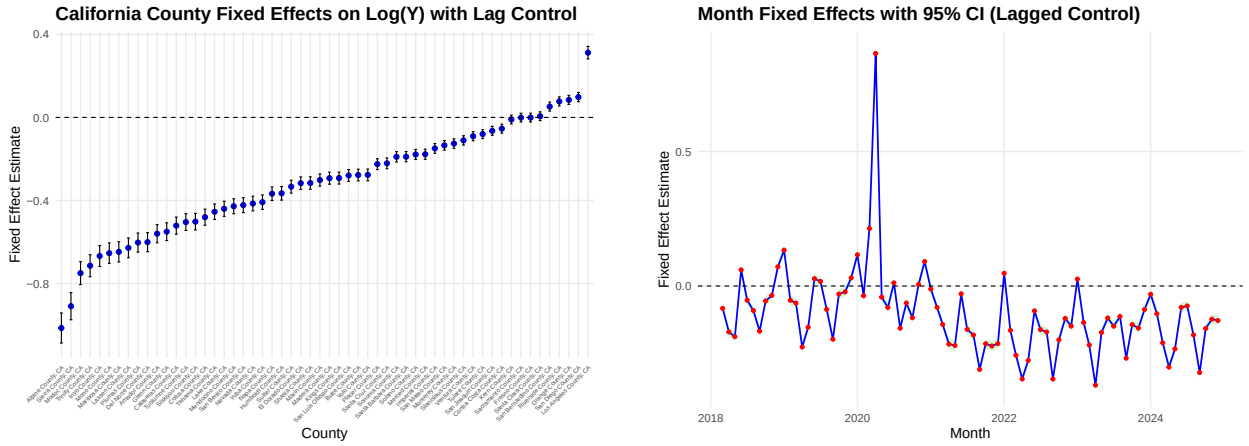
Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4a and Figure 4b visualize the estimated fixed effects for California counties and monthly time periods, respectively, under the two-way fixed effects model with lagged outcome control. The estimated fixed effects for California counties capture persistent differences in unemployment levels across locations after controlling for treatment and time. Notably, counties such as Los Angeles, San Diego, and Orange exhibit strongly positive fixed effects, indicating systematically higher baseline unemployment levels. This is consistent with their large population size, urban density, and diverse labor markets with greater exposure to policy shocks and cyclical industries. In contrast, rural and sparsely populated counties like Alpine and Sierra display strongly negative fixed effects, suggesting structurally lower unemployment. These areas often have limited economic complexity and more stable employment patterns driven by public sector or seasonal industries.

Furthermore, the plot reveals that the confidence intervals for counties with lower fixed effects tend to be wider, while those for counties with higher fixed effects are relatively narrow. This can be explained by differences in data availability and sample size: large urban counties provide more observations and thus allow for more precise estimation, while small counties exhibit greater

variability due to limited data. Regarding the big picture, the majority of county-level fixed effects are estimated with high precision, as reflected by the generally narrow confidence intervals.

The estimated month fixed effects also align closely with the time-series trends observed in our data description. The fitted values capture both the long-term upward drift in unemployment and the dramatic positive spike during the early COVID-19 period, demonstrating that the model successfully absorbs key temporal shocks. Importantly, the confidence intervals around each monthly effect are extremely tight, indicating that all month fixed effects are highly statistically significant. This reinforces the role of time fixed effects in controlling for economy-wide shocks that influence unemployment levels uniformly across counties.



(a) County Fixed Effects in California

(b) Month Fixed Effects

Figure 4: Fixed Effects Visualizations

6.2 Heterogeneity Analysis

To assess whether the policy impact differs by the baseline level of unemployment, we conduct a heterogeneity analysis by interacting the treatment effect with a binary indicator of high unemployment. Specifically, we define a county as "high unemployment" if its average unemployment rate prior to January 2021 exceeds a certain threshold, and "low unemployment" otherwise. We consider two thresholding approaches: (1) a pooled median across both California and New York

counties, and (2) state-specific medians computed separately within each state.

The heterogeneity is tested through the inclusion of a triple interaction term between the treatment group indicator, post-treatment indicator, and the high-unemployment indicator. Under the Rubin Causal Model framework, this interaction term captures the difference in treatment effects between high- and low-unemployment counties. A statistically significant coefficient on this triple interaction term would indicate that the policy effect varies meaningfully with baseline unemployment conditions.

Table 3 reports the results from both specifications. In both models, the coefficient on the baseline treatment effect (Post × Treatment) is positive and statistically significant, indicating that California's low-unemployment counties experienced an approximate 2.3% increase in unemployment following the policy relative to New York counties. However, the triple interaction terms are small in magnitude and statistically insignificant (p-values of 0.675 and 0.993, respectively). This suggests that the policy's effect does not differ meaningfully between high- and low-unemployment counties, regardless of how the baseline unemployment classification is constructed.

Table 3: Heterogeneous DID Estimates by Unemployment Level

	Cross-State Median	**State-Specific Median**
Intercept	0.921^{***}	0.920^{***}
	(0.033)	(0.033)
Post	0.107^{***}	0.106^{***}
	(0.009)	(0.009)
Treatment	0.231^{***}	2.307^{***}
	(0.014)	(0.013)
log(Y_{it-1})	0.854^{***}	0.854^{***}
	(0.006)	(0.006)
High Unemployment	0.408^{***}	4.083^{***}
	(0.018)	(0.018)
Post × Treatment	0.022^{***}	0.023^{***}
	(0.004)	(0.003)
Post × Treatment × High Unemployment	0.00177	−0.00004
	(0.0042)	(0.0041)
Observations	10,080	10,080
R^2	0.9983	0.9982
Adjusted R^2	0.9982	0.9982
F-statistic	27623.71	27623.22

Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

6.3 Robust Checks

To formally test the parallel trends assumption, we estimate a series of regressions in which log unemployment is regressed on state-specific linear, quadratic, and cubic time trends. Specifically, we interact the New York dummy with a time index, its square, and its cube to capture potential differences in linear and nonlinear pre-trends. Table 4 reports the results. In each specification, the interaction terms between the New York dummy and the time polynomials are statistically insignificant. Importantly, the insignificance of these interaction terms suggests that differences between states in unemployment do not evolve systematically with time prior to the treatment. In other words, there is no evidence of differential time trends across states that could bias the DID estimates. These results provide supportive evidence that the parallel trends assumption holds in the pre-treatment period.

Table 4: Statistical Tests for Pre-Treatment Parallel Trends

	$\log(Y_{2018\text{-}2020})$	$\log(Y_{2018\text{-}2020})$, Quad.	$\log(Y_{2018\text{-}2020})$, Cubic
New York × Time Index	-0.000009	-0.00029	-0.00074
	(0.00026)	(0.00083)	(0.0020)
New York × Time Index^2		0.0000003	0.000001
		(0.0000008)	(0.0000044)
New York × Time Index^3			-0.0000000007
			(0.000000003)
Observations	72	72	72
R^2	0.658	0.768	0.779
Adjusted R^2	0.643	0.750	0.7548
F-statistic	43.550	43.67	32.22

Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Regressions are done on state-level data.

In this section, to further support the validity of our identification strategy, we also conduct a series of robustness checks and placebo tests. Specifically, we test whether our findings hold under

alternative outcome variables, placebo treatment dates, and alternative policy shock assumptions.

We begin by replacing the outcome variable with log population instead of log unemployment to examine whether our main findings are driven by demographic shifts. Since population levels are unlikely to react to minimum wage policy changes in the short run, this serves as a valid placebo outcome. In addition to the baseline specification, we estimate an alternative model that includes lagged log population and two-way fixed effects. As shown in Figure 5, the specification yields statistically insignificant treatment effects, reinforcing that our baseline estimates are not confounded by underlying population dynamics.

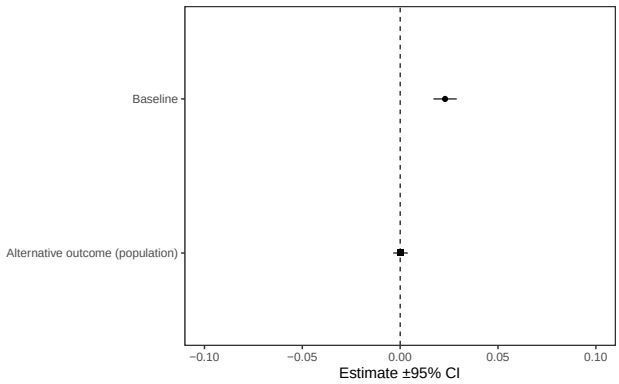


Figure 5: Alternative Outcome

Next, we perform placebo tests by shifting the policy intervention date. We estimate the same model using three alternative treatment timings: January 2019, January 2020, and January 2022. Each of these placebo dates serves a distinct purpose. The 2019 and 2020 specifications test for potential pre-treatment shocks or anticipation effects, while the 2022 specification checks for spurious treatment effects after the actual policy implementation. As shown in figure 6, all placebo coefficients are close to zero and statistically insignificant. These results suggest that the estimated treatment effect in the baseline model is not driven by timing artifacts or unrelated shocks, lending support to the internal validity of our DID design.

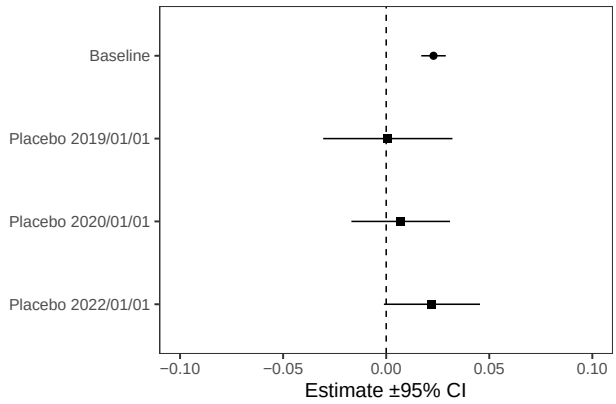


Figure 6: Placebo Dates Check

Finally, we use the onset of COVID-19 as an alternative source of policy variation. Specifically, we set March 2020—the beginning of widespread lockdown measures in the United States—as the placebo treatment date. This allows us to test whether the observed treatment effect in our baseline model could instead be driven by confounding shocks related to the pandemic.

We acknowledge two important limitations in our sector-specific placebo analysis. First, due to data constraints, we focus on log employment rather than unemployment. If sector-level unemployment data were available, we could have directly tested heterogeneous effects in unemployment, providing stronger identification. Second, the analysis is based on state-level panel data for only two units—California and New York—which raises concerns for inference. While the model reports standard errors, these rely on strong assumptions, such as independently and identically distributed residuals across time. In practice, such assumptions are unlikely to hold, especially in a DID setting with a small number of aggregated units. As a result, the reported standard errors may underestimate true uncertainty. Therefore, we interpret these sectoral placebo results descriptively rather than making strong statistical claims.

As shown in Table 5, the DID estimates for both sectors remain statistically insignificant. These results suggest that the baseline treatment effect is not confounded by employment shocks caused by the COVID-19 outbreak. In a descriptive sense, this robustness check further supports the credibility of our identification strategy.

Table 5: Placebo Tests Using Sector-Specific Employment Outcomes

	Tech Sector	**Food Industry**
Intercept	1.2007 (0.6395)	5.5058 (1.1549)
Treatment	0.0935 (0.0494)	0.2850 (0.0608)
Post	0.0223 (0.0115)	-0.0618 (0.0441)
log($Y_{i,t-1}$)t	0.9004 (0.0521)	0.5680 (0.0907)
Treatment × Post	-0.0034 (0.0039)	0.0172 (0.0147)
Observations	172	172
R^2	0.9997	0.9951
Adjusted R^2	0.9994	0.9898
F-statistic	3379.89	189.95

Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

In summary, all kinds of robustness checks confirm that our main conclusions do not depend on specific modeling choices, outcome definitions, or sample timing. The insignificant results under alternative outcomes and placebo timings indicate that our baseline estimates are unlikely to be driven by unobserved confounding factors.

6.4 Potential Problems

A major limitation of our identification strategy is the potential presence of simultaneous state-level shocks that differentially impacted California and New York. Around January 2021, California extended its stay-at-home orders in response to rising COVID-19 cases, while New York relaxed similar restrictions earlier. This divergence in containment policy could have suppressed labor demand more heavily in California, introducing a downward bias in our estimate of the minimum wage effect. In parallel, California's unemployment insurance system faced severe administrative challenges, including processing delays, fraud investigations, and benefit backlogs. These disruptions may have temporarily inflated the reported number of unemployed individuals in California

relative to New York, artificially elevating the baseline against which post-treatment employment recovery is measured. This would likely bias the treatment effect estimate upward. Together, these two factors—public health restrictions and administrative failures—introduce offsetting biases in opposite directions. However, since neither shock is orthogonal to the timing of the minimum wage increase, they complicate the causal interpretation of our estimates. The net bias is difficult to sign ex ante, but its presence underscores the importance of robustness checks and careful contextualization of results.

A second limitation relates to the level of aggregation in the outcome variable. The analysis uses total county-level unemployment rather than sector-specific measures. This aggregation may mask heterogeneous treatment effects across industries, particularly in low-wage sectors more directly affected by the minimum wage increase (e.g., retail, hospitality, food services). The lack of disaggregated data prevents a more targeted estimation and may bias estimates toward zero if treatment effects are concentrated in particular subgroups.

If we had access to an ideal dataset, we would incorporate time-varying county-level covariates, such as demographic structure, industry composition, and local policy activity. Including such controls would allow us to better address potential omitted variable bias, particularly when such factors evolve differently across treatment and control units over time. In doing so, we could more accurately isolate the causal impact of the minimum wage increase by netting out confounding trends correlated with both treatment assignment and labor market outcomes.

Moreover, with more granular data at the individual or firm level, we could explicitly model treatment effect heterogeneity and implement more credible robustness checks. For example, panel data would enable us to exploit within-unit variation and perform event-study analyses to detect dynamic responses to the policy. It would also facilitate falsification exercises by comparing affected and unaffected subgroups (e.g., high-wage vs. low-wage sectors).

7 Conclusion

This study evaluates the causal impact of California's 2021 minimum wage increase on county-level unemployment using a Difference-in-Differences framework with county and time fixed effects and a lagged dependent variable. Drawing on panel data from California and New York counties between 2018 and 2025, our empirical design accounts for both persistent regional heterogeneity and dynamic labor market adjustment.

To address the first research question, our main specification estimates that the policy led to a statistically significant 2.3% increase in unemployment in California relative to New York. This effect remains robust after controlling for fixed effects, serial correlation, and time-specific shocks. Additional robustness checks—including parallel trend checks, placebo tests and alternative outcome analysis—support the credibility of this finding.

To answer the second research question, we examine whether the policy impact differs across counties with varying pre-policy unemployment levels. We classify counties into high and low unemployment groups based on both pooled and state-specific medians and implement a triple interaction framework. In all specifications, the interaction terms are statistically insignificant, indicating no evidence of heterogeneous effects. The policy's impact on unemployment appears consistent across counties, regardless of their initial labor market conditions.

In summary, our findings support the competitive labor market theory's prediction that raising the minimum wage above the level at which the market clears can result in lower employment due to higher labor costs. Although it is consistent across model choices and subgroup definitions, the estimated employment effect is moderate. These results imply that, even in the event of a post-pandemic labor market recovery, raising the minimum wage may come with actual and widely dispersed costs.

8 References

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